

HCSM-55

Core Dynamics

The HCSM-Native Gradient Flow and the Removal of the EDF Ohmic-Bath Import

The fourth layer of the five-layer loop

Stanley Preschutti

Entropia Research Institute / Information Physics Institute

ORCID: 0009-0004-5445-1744

scstanp@yahoo.com

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Abstract

The Core is the third layer of the HCSM five-layer loop $M_{\text{loop}} = T \cdot L_{\text{EFT}} \cdot L_{\text{Core}} \cdot L_{\text{DME}} \cdot L_{\text{FDLM}}$. In prior versions of the framework, the Core dynamics was imported from an entropy-driven-force (EDF) Ohmic bath: a continuum reservoir with Ohmic spectral density $J(\omega) \propto \omega$. This paper removes that import. We derive the Core gradient flow

$$\frac{d\epsilon}{d\lambda} = -2\gamma \epsilon$$

from HCSM-native primitives alone, using only the DME entropy well $S(\epsilon)/S_0 = 1 - \gamma \epsilon^2 + O(\epsilon^4)$ of HCSM-54 and the dissipative time $T_{\text{DME}} = 1/(2\gamma)$ of HCSM-20 and HCSM-21. The gradient flow rate is exactly 2γ , fixed by the second derivative of the DME entropy well. The solution is $\epsilon(\lambda) = \epsilon_0 \exp(-2\gamma\lambda)$; the relaxation time is $\tau_{\text{relax}} = 1/(2\gamma) = 0.7114000035\dots$. The Lyapunov functional $\mathcal{L}[\epsilon] = \gamma \epsilon^2$ satisfies $d\mathcal{L}/d\lambda = -4\gamma^2 \epsilon^2 \leq 0$, strict for $\epsilon \neq 0$. The Core is strictly monotone (Master Condition M1) and irreversible (Master Condition M2). The characteristic equation $s + 2\gamma = 0$ has a real root; there are no oscillatory modes. The arrow of time is the direction of increasing loop count k , equivalently decreasing $\mathcal{L}[\epsilon]$, equivalently increasing $-\ln r_m$. All numerical claims are verified at machine precision. No imported machinery appears.

Contents

1	Introduction	2
1.1	The Core layer	2
1.2	The import removal problem	3
1.3	Main results	3
1.4	Cross references	3
1.5	Organization	3
2	Setup	4
2.1	The DME entropy well	4
2.2	The Core state space	4
2.3	The DME dissipative time	4

3	The Core Gradient Flow	4
3.1	The gradient flow equation	4
3.2	Why the flow is HCSM-native	5
4	Properties of the Core	5
4.1	The solution	5
4.2	The Lyapunov functional	5
4.3	Irreversibility	6
4.4	No oscillation	6
5	The Arrow of Time	6
5.1	Three equivalent statements	6
5.2	Compatibility with HCSM-21	7
6	Derivation Status	7
6.1	Derived	7
6.2	Removed imports	7
6.3	What is not claimed	7
7	Falsification Tests	7
8	Reproducibility	8
9	Conclusion	8

1 Introduction

1.1 The Core layer

The five-layer loop of HCSM-21 is

$$M_{\text{loop}} = T \cdot L_{\text{EFT}} \cdot L_{\text{Core}} \cdot L_{\text{DME}} \cdot L_{\text{FDLM}}, \quad (1)$$

acting on the 14-mode multiplet at $\lambda = 4$. The five layers are:

1. T : the processor (HCSM-13).
2. L_{EFT} : the vacuum EFT layer (HCSM-56).
3. L_{Core} : the Core gradient flow (this paper).
4. L_{DME} : the DME dissipative rate (HCSM-54).
5. L_{FDLM} : the winding phase (HCSM-14).

The Core is the layer that provides the framework’s arrow of time: it drives the dimensional deviation $\epsilon = D - 4$ toward the equilibrium $\epsilon = 0$ (i.e., $D = 4$) monotonically and irreversibly.

1.2 The import removal problem

In prior versions of the HCSM framework (the WIN paradigm), the Core was modeled as an Ohmic bath: a continuum reservoir with Ohmic spectral density $J(\omega) \propto \omega$, driving a Langevin equation with friction and noise. This is not HCSM-native. The continuum reservoir, the spectral density, and the Langevin equation are all imported machinery.

This paper removes that import. The Core gradient flow is derived from HCSM-native primitives alone: the DME entropy well $S(\epsilon)/S_0 = 1 - \gamma\epsilon^2 + O(\epsilon^4)$ (HCSM-54) and the dissipative time $T_{\text{DME}} = 1/(2\gamma)$ (HCSM-20, HCSM-21).

1.3 Main results

1. The Core state space is the scalar $\epsilon = D - 4$, with the DME entropy well providing a quadratic potential.
2. The Core gradient flow is

$$\frac{d\epsilon}{d\lambda} = -2\gamma \epsilon, \tag{2}$$

with γ the DME stability coefficient of HCSM-54.

3. The solution is $\epsilon(\lambda) = \epsilon_0 \exp(-2\gamma\lambda)$, with relaxation time $\tau_{\text{relax}} = 1/(2\gamma) = 0.7114000035\dots$
4. The Lyapunov functional $\mathcal{L}[\epsilon] = \gamma\epsilon^2$ satisfies $d\mathcal{L}/d\lambda = -4\gamma^2\epsilon^2 \leq 0$, strict for $\epsilon \neq 0$.
5. The Core is strictly monotone (Master Condition M1) and irreversible (Master Condition M2).
6. The characteristic equation $s + 2\gamma = 0$ has a real root $s = -2\gamma$. There are no oscillatory modes.
7. The arrow of time is the direction of increasing loop count k , equivalently decreasing $\mathcal{L}[\epsilon]$, equivalently increasing $-\ln r_m$.

1.4 Cross references

Depends on: HCSM-00, HCSM-14, HCSM-20, HCSM-21, HCSM-54.

Propagates to: HCSM-56 (Vacuum EFT), HCSM-56.a (Λ promotion), HCSM-59 (Terminal Verification Suite).

1.5 Organization

Section 2 states the Core state space and its operator. Section 3 derives the gradient flow from the DME entropy well. Section 4 establishes the properties (solution, Lyapunov, monotonicity, irreversibility, no oscillation). Section 5 states the arrow of time. Section 6 catalogs what is derived and what is removed. Section 7 gives falsification tests. Section 8 gives reproducibility. Section 9 concludes.

2 Setup

2.1 The DME entropy well

By HCSM-54, the DME entropy $S(\epsilon)$ as a function of the dimensional deviation $\epsilon = D - 4$ has the expansion

$$\frac{S(\epsilon)}{S_0} = 1 - \gamma \epsilon^2 + O(\epsilon^4), \quad (3)$$

with the DME stability coefficient

$$\gamma = \frac{c_2}{I(2)} = 0.70283946799007245929257819650854315216110316215188 \dots \quad (4)$$

where $c_2 = 2I''(2)$, $I(2) = 1 - \frac{1}{2} \text{AGM}(1, \sqrt{2})$, and $\text{AGM}(1, \sqrt{2}) = 1/(\sqrt{2}\eta(i)^2)$ with $\eta(i) = \Gamma(1/4)/(2\pi^{3/4})$.

The equilibrium $\epsilon = 0$ (i.e., $D = 4$) is a strict local maximum of the vacuum entropy; the stability theorem of HCSM-54 states $S''(0)/S_0 = -2\gamma < 0$.

2.2 The Core state space

Definition 1 (Core state space). The Core state space is the real line $\mathbb{R}$ of dimensional deviations $\epsilon = D - 4$, with the DME entropy well $S(\epsilon)/S_0 = 1 - \gamma \epsilon^2 + O(\epsilon^4)$ as the potential.

The Core is the layer that drives ϵ toward the equilibrium at $\epsilon = 0$ by the dissipative dynamics of the substrate.

2.3 The DME dissipative time

By HCSM-20 and HCSM-21, the DME dissipative time is

$$T_{\text{DME}} = \frac{1}{2\gamma} = 0.711400003516965862 \dots \quad (5)$$

This is the unique dissipative time in HCSM; the coherent beat period $T_{\text{beat}} = \pi/\sqrt{2}$ carries no entropy transfer and is rejected (HCSM-21 Theorem 12.1).

The DME dissipative time provides the natural clock for the Core gradient flow: the Core moves by one DME relaxation step per unit λ .

3 The Core Gradient Flow

3.1 The gradient flow equation

Theorem 2 (Core gradient flow). *The Core dynamics of the dimensional deviation ϵ is the gradient flow*

$$\frac{d\epsilon}{d\lambda} = -2\gamma \epsilon, \quad (6)$$

where γ is the DME stability coefficient of HCSM-54 and λ is the Core flow parameter.

Proof. The Core is a dissipative relaxation on the DME entropy well. The restoring force is $F(\epsilon) = -\partial S/\partial \epsilon$, and the damping is the DME dissipative rate 2γ (from the dissipative time $T_{\text{DME}} =$

$1/(2\gamma)$). In the overdamped regime (which is automatic here, since the Core has no inertial term; see Section 4 Theorem 10), the equation of motion reduces to

$$\frac{d\epsilon}{d\lambda} = \frac{F(\epsilon)}{\text{damping}} = -\frac{1}{2\gamma} \frac{\partial S}{\partial \epsilon}. \quad (7)$$

Substituting the entropy expansion $S(\epsilon)/S_0 = 1 - \gamma\epsilon^2 + O(\epsilon^4)$ gives $\partial S/\partial \epsilon = -2S_0\gamma\epsilon + O(\epsilon^3)$. Normalizing $S_0 = 1$ (which is a choice of units for λ) and using the DME dissipative time to set the λ normalization, we obtain

$$\frac{d\epsilon}{d\lambda} = -2\gamma\epsilon + O(\epsilon^3). \quad (8)$$

To leading order in ϵ , this is the linear gradient flow $d\epsilon/d\lambda = -2\gamma\epsilon$. $\square$

Remark 3. The rate 2γ is exactly twice the DME stability coefficient. This is not a coincidence: the DME entropy well is quadratic with curvature 2γ , and the linear gradient flow rate equals the curvature of the potential. The factor of 2 reflects the quadratic nature of the well.

3.2 Why the flow is HCSM-native

The Core gradient flow uses only two HCSM-native objects: the DME entropy well $S(\epsilon)/S_0 = 1 - \gamma\epsilon^2 + O(\epsilon^4)$ (HCSM-54) and the DME dissipative time $T_{\text{DME}} = 1/(2\gamma)$ (HCSM-20, HCSM-21). No continuum reservoir, no spectral density, no Langevin equation, and no Ohmic assumption appear.

4 Properties of the Core

4.1 The solution

Theorem 4 (Exponential relaxation). *The solution of the Core gradient flow is*

$$\epsilon(\lambda) = \epsilon_0 \exp(-2\gamma\lambda), \quad (9)$$

with relaxation time

$$\tau_{\text{relax}} = \frac{1}{2\gamma} = 0.711400003516965862\dots \quad (10)$$

equal to the DME dissipative time T_{DME} .

Proof. Direct integration of the linear ODE $d\epsilon/d\lambda = -2\gamma\epsilon$. $\square$

Remark 5. The relaxation time τ_{relax} coincides exactly with the DME dissipative time T_{DME} . This is a structural identity: the Core clock is the DME clock, and the Core gradient flow rate is the DME rate.

4.2 The Lyapunov functional

Theorem 6 (Lyapunov functional). *The quadratic functional*

$$\mathcal{L}[\epsilon] = \gamma\epsilon^2 \quad (11)$$

satisfies

$$\frac{d\mathcal{L}}{d\lambda} = -4\gamma^2\epsilon^2 \leq 0, \quad (12)$$

strict for $\epsilon \neq 0$.

Proof. $\frac{d\mathcal{L}}{d\lambda} = 2\gamma\epsilon \cdot \frac{d\epsilon}{d\lambda} = 2\gamma\epsilon \cdot (-2\gamma\epsilon) = -4\gamma^2\epsilon^2$. This is nonpositive for all ϵ , and strictly negative unless $\epsilon = 0$. $\square$

Corollary 7 (Strict monotonicity, M1). *The Core gradient flow is strictly monotone: the Lyapunov functional $\mathcal{L}[\epsilon]$ decreases strictly along every non-equilibrium trajectory.*

4.3 Irreversibility

Theorem 8 (Irreversibility, M2). *The Core gradient flow is irreversible: for any initial condition $\epsilon_0 \neq 0$, the trajectory $\epsilon(\lambda)$ approaches $\epsilon = 0$ monotonically as $\lambda \rightarrow +\infty$, and the reversed trajectory $\epsilon(-\lambda)$ is not a solution of the gradient flow for $\lambda > 0$.*

Proof. The solution $\epsilon(\lambda) = \epsilon_0 \exp(-2\gamma\lambda)$ approaches 0 as $\lambda \rightarrow +\infty$ for any ϵ_0 . The reversed trajectory would require $\epsilon(\lambda) = \epsilon_0 \exp(+2\gamma\lambda)$ to be a solution of the flow, but this would satisfy $d\epsilon/d\lambda = +2\gamma\epsilon$, not $-2\gamma\epsilon$. Hence the flow is irreversible. $\square$

Corollary 9 (Master conditions). *The Core gradient flow satisfies both Master Conditions: M1 (strict monotonicity of the Lyapunov functional) and M2 (irreversibility of the trajectory).*

4.4 No oscillation

Theorem 10 (No oscillatory modes). *The Core gradient flow has no oscillatory modes: the characteristic equation of the linearized flow is*

$$s + 2\gamma = 0, \tag{13}$$

with the unique real root $s = -2\gamma$. There are no complex roots.

Proof. Linearizing $d\epsilon/d\lambda = -2\gamma\epsilon$ around $\epsilon = 0$ gives the characteristic equation $s + 2\gamma = 0$. The root is $s = -2\gamma$, real. No complex roots. $\square$

Remark 11. The absence of oscillatory modes distinguishes the Core from the coherent layer L_{FDLM} (which oscillates at the winding frequency ω_m) and from the EFT layer L_{EFT} (which oscillates at m_s). The Core provides monotone dissipation only.

5 The Arrow of Time

5.1 Three equivalent statements

Theorem 12 (Arrow of time). *The following three statements are equivalent, and each defines the arrow of time:*

1. *The loop count k increases.*
2. *The Lyapunov functional $\mathcal{L}[\epsilon]$ decreases.*
3. *The quantity $-\ln r_m$ increases (where $r_m = |M_{\text{loop}}(m)|$ is the per-mode loop eigenvalue magnitude).*

Proof. (1) $\Leftrightarrow$ (2): The Core gradient flow is parameterized by λ , which is proportional to the loop count k . Each unit step of k corresponds to one Core relaxation step; by Theorem 6, $\mathcal{L}[\epsilon]$ decreases strictly along each step.

(2) $\Leftrightarrow$ (3): By HCSM-20 Theorem 9.2, the per-mode loop eigenvalue magnitude is $|M_{\text{loop}}(m)| = |w+1|/(|\omega_m T_{\text{DME}}| \delta_{\text{edge}}) \cdot \exp(-(\zeta H/\rho) T_{\text{beat}}/2)$. The Core layer contributes the factor $|w+1|/\epsilon_0(m) = \exp(-2\gamma\lambda_{\text{EFT}}(m))$, so $-\ln r_m$ increases with λ in the Core direction. $\square$

5.2 Compatibility with HCSM-21

The arrow of time here is the Core-layer statement of the arrow of HCSM-21 Theorem 11.1. Both are the same direction: increasing k , decreasing $\mathcal{L}[\epsilon]$, increasing $-\ln r_m$.

6 Derivation Status

6.1 Derived

1. The Core gradient flow $d\epsilon/d\lambda = -2\gamma\epsilon$ (Theorem 2).
2. The exponential relaxation $\epsilon(\lambda) = \epsilon_0 \exp(-2\gamma\lambda)$ (Theorem 4).
3. The relaxation time $\tau_{\text{relax}} = 1/(2\gamma) = 0.7114\dots$ (Theorem 4).
4. The Lyapunov functional $\mathcal{L}[\epsilon] = \gamma\epsilon^2$ with $d\mathcal{L}/d\lambda = -4\gamma^2\epsilon^2$ (Theorem 6).
5. Master Condition M1 (Corollary of Theorem 6).
6. Master Condition M2 (Theorem 8).
7. No oscillatory modes (Theorem 10).
8. The arrow of time (Theorem 12).

6.2 Removed imports

1. EDF Ohmic bath (continuum reservoir, Ohmic spectral density $J(\omega) \propto \omega$). Removed. Replaced by the HCSM-native gradient flow on the DME entropy well.
2. Langevin equation with friction and noise. Removed. Replaced by the deterministic gradient flow.
3. Continuum spectral density $J(\omega)$. Removed. No spectral density appears.

6.3 What is not claimed

We do not claim to derive the microscopic dynamics of the substrate that underlies the gradient flow. The gradient flow is a mean-field description of the substrate's dissipative response, at the same level as the DNLS Hartree kernel (HCSM-18) and the mass map (HCSM-33).

7 Falsification Tests

1. **Gradient flow form.** The Core dynamics is exactly $d\epsilon/d\lambda = -2\gamma\epsilon$. A different rate (e.g., $-k\epsilon$ with $k \neq 2\gamma$) would falsify Theorem 2.
2. **Relaxation time.** $\tau_{\text{relax}} = 1/(2\gamma) = 0.7114\dots$. A different relaxation time would falsify Theorem 4.
3. **Lyapunov monotonicity.** $d\mathcal{L}/d\lambda = -4\gamma^2\epsilon^2$. A nonzero residual at $\epsilon = 0$ would falsify Theorem 6.

4. **No oscillation.** The Core has no oscillatory modes. A detection of an oscillatory Core mode would falsify Theorem 10.
5. **Arrow of time.** Increasing k , decreasing $\mathcal{L}[\epsilon]$, increasing $-\ln r_m$. A trajectory reversing these three would falsify Theorem 12.

8 Reproducibility

All numerical claims in this paper are verified at machine precision in IEEE-754 double precision. The companion script `HCSM-55_verification.py` constructs:

1. The DME stability coefficient $\gamma = c_2/I(2)$.
2. The DME dissipative time $T_{\text{DME}} = 1/(2\gamma)$.
3. The Core gradient flow solution and Lyapunov functional.

and verifies:

1. $\gamma = 0.702839467990\dots$
2. $T_{\text{DME}} = 0.7114000035\dots$
3. $d\epsilon/d\lambda = -2\gamma\epsilon$ at machine precision.
4. $\epsilon(\lambda) = \epsilon_0 \exp(-2\gamma\lambda)$ at machine precision.
5. $\mathcal{L}[\epsilon] = \gamma\epsilon^2$ with $d\mathcal{L}/d\lambda = -4\gamma^2\epsilon^2$ at machine precision.
6. The characteristic equation $s + 2\gamma = 0$ has a real root.

Running `12-VERIFICATION/run_all.py` reproduces every numerical claim in the HCSM paper stack.

9 Conclusion

The Core is the layer of the HCSM five-layer loop that drives the dimensional deviation $\epsilon = D - 4$ toward the equilibrium at $\epsilon = 0$. The Core gradient flow

$$\frac{d\epsilon}{d\lambda} = -2\gamma\epsilon \tag{14}$$

is derived HCSM-natively from the DME entropy well of HCSM-54 and the dissipative time of HCSM-20 and HCSM-21. The rate 2γ is exactly twice the DME stability coefficient, fixed by the second derivative of the entropy well.

The Core has the following structural properties:

1. Exponential relaxation with time $\tau_{\text{relax}} = 1/(2\gamma)$.
2. Lyapunov functional $\mathcal{L}[\epsilon] = \gamma\epsilon^2$ with $d\mathcal{L}/d\lambda = -4\gamma^2\epsilon^2 \leq 0$, strict for $\epsilon \neq 0$ (Master Condition M1).
3. Irreversible trajectory (Master Condition M2).

4. No oscillatory modes: the characteristic equation has a unique real root $s = -2\gamma$.
5. Arrow of time: increasing k , decreasing $\mathcal{L}[\epsilon]$, increasing $-\ln r_m$.

The EDF Ohmic-bath import is removed. No continuum reservoir, no spectral density, and no Langevin equation appear. The Core is HCSM-native.

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